Introduction to Demographic Methods

(2: Mortality)

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Introduction to Demographic Methods

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1 Introduction. Why study Demography? . . . Git, version control, course materials GitHub
2 Mortality
3 Fertility
4 Population projections
5 Migration
~~6 Kinship~~
7 The future, and Digital and Computational Demography
8 Wrap up + Description of examination
9 Final examination: 3 min presentation to share your selected topic + 2 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Few points

- Important difference between:
  - Rate (occurrence / person years lived, has time in the denominator)
  - Ratio (two values divided and compared)
  - Probability (occurrence / previous occurrences, does not have time in the denominator)
- Based on Wachter, 2014, refer to slides 18-27 of Ernesto Amaral's lecture 02
- Using mortality measures for a cohort is easier as it allows us to only look at mortality
- No one can enter this group and population after they are born into the cohort.
- They can only die and exit this cohort.

Switch to Ernesto Amaral's slides: Lecture03

Hands-on part

Needed installations (or checking requirements)

For today’s hands-on part, see the Readme file under “03_MonicaALabs” folder and download the requested files under “data” folder.

We will go over the notebook called “2_mortality_decomp_edits.qmd” and see what it does. You need to activate Renv and install R packages listed in the first code block of the notebook, i.e., "tidyverse, here, readxl, janitor" (please check the notebook for the latest list).

Needed: R, RStudio, and Renv environment.

Where to learn basics of R and Python?

To start with Python

Check this website first:
https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:
https://github.com/vtraag/intro-python

Or this one by Data Carpentry:
https://datacarpentry.org/python-ecology-lesson/

To start with R

Check this website first:
https://www.w3schools.com/r/default.asp

This "very short introduction to R" is a good start:
https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf

Or this course by Data Carpentry:
https://datacarpentry.org/R-genomics/index.html

Reading materials for today

Key reading material, Chapters 3 and 7

- Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.

Additional reading materials

- Alexander, M. (2026). Advanced Data Analysis course in Demographic Methods. Retrieved April 8, 2026, from https://github.com/MJAlexander/soc6708
- Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the "key reading material".
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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References

Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.